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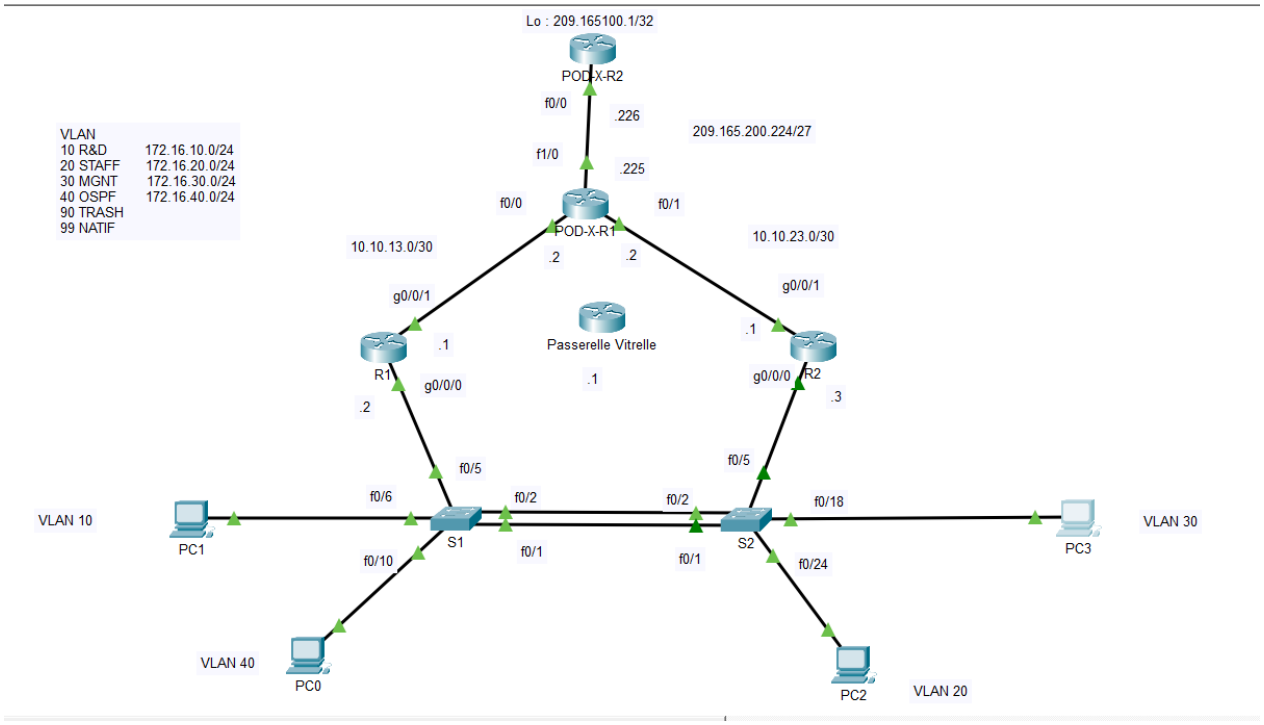
Labo NAT



ip r sur pc1 :

```
root@poda-1:~# ip r
default via 172.16.10.1 dev eth0
172.16.10.0/24 dev eth0 proto kernel scope link src 172.16.10.22
root@poda-1:~#
```

ip a sur pc1 :

```
root@poda-1:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0@if20: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group
default qlen 1000
    link/ether 2e:0a:66:2b:82:8b brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.16.10.22/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet 172.16.10.10/24 scope global secondary eth0
       valid_lft forever preferred_lft forever
root@poda-1:~#
```

ip r sur pc2 :

```
root@poda-2:~# ip r
default via 172.16.20.1 dev eth0
172.16.20.0/24 dev eth0 proto kernel scope link src 172.16.20.10
root@poda-2:~#
```

ip a sur pc2 :

```
root@poda-2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0@if32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group
default qlen 1000
    link/ether 9e:d3:d7:33:17:52 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.16.20.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::9cd3:d7ff:fe33:1752/64 scope link
       valid_lft forever preferred_lft forever
root@poda-2:~#
```

ip r sur pc3 :

```
root@poda-3:~# ip r
default via 172.16.30.1 dev eth0
172.16.30.0/24 dev eth0 proto kernel scope link src 172.16.30.10
root@poda-3:~#
```

ip a sur pc3 :

```
root@poda-3:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0@if36: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group
default qlen 1000
    link/ether 96:b6:2f:97:a1:7c brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.16.30.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::94b6:2fff:fe97:a17c/64 scope link
       valid_lft forever preferred_lft forever
root@poda-3:~#
```

ping depuis PC1 vers PC2

```
root@poda-1:~# ping 172.16.20.10
PING 172.16.20.10 (172.16.20.10) 56(84) bytes of data.
64 bytes from 172.16.20.10: icmp_seq=1 ttl=63 time=0.735 ms
64 bytes from 172.16.20.10: icmp_seq=2 ttl=63 time=0.741 ms
64 bytes from 172.16.20.10: icmp_seq=3 ttl=63 time=0.746 ms
64 bytes from 172.16.20.10: icmp_seq=4 ttl=63 time=0.745 ms
64 bytes from 172.16.20.10: icmp_seq=5 ttl=63 time=0.749 ms
64 bytes from 172.16.20.10: icmp_seq=6 ttl=63 time=0.733 ms
^C
--- 172.16.20.10 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5128ms
rtt min/avg/max/mdev = 0.733/0.741/0.749/0.005 ms
root@poda-1:~#
```

ping depuis PC1 vers PC3

```
root@poda-1:~# ping 172.16.30.10
PING 172.16.30.10 (172.16.30.10) 56(84) bytes of data.
64 bytes from 172.16.30.10: icmp_seq=2 ttl=63 time=0.763 ms
64 bytes from 172.16.30.10: icmp_seq=3 ttl=63 time=0.517 ms
64 bytes from 172.16.30.10: icmp_seq=4 ttl=63 time=0.744 ms
64 bytes from 172.16.30.10: icmp_seq=5 ttl=63 time=0.719 ms
64 bytes from 172.16.30.10: icmp_seq=6 ttl=63 time=0.743 ms
^C
--- 172.16.30.10 ping statistics ---
6 packets transmitted, 5 received, 16.6667% packet loss, time 5106ms
rtt min/avg/max/mdev = 0.517/0.697/0.763/0.091 ms
root@poda-1:~#
```

ping depuis PC1 vers S1

```
root@poda-1:~# ping 172.16.30.11
PING 172.16.30.11 (172.16.30.11) 56(84) bytes of data.
64 bytes from 172.16.30.11: icmp_seq=1 ttl=254 time=0.728 ms
64 bytes from 172.16.30.11: icmp_seq=2 ttl=254 time=0.796 ms
64 bytes from 172.16.30.11: icmp_seq=3 ttl=254 time=0.774 ms
64 bytes from 172.16.30.11: icmp_seq=4 ttl=254 time=1.66 ms
64 bytes from 172.16.30.11: icmp_seq=5 ttl=254 time=0.770 ms
^C
--- 172.16.30.11 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4056ms
rtt min/avg/max/mdev = 0.728/0.945/1.657/0.356 ms
root@poda-1:~#
```

ping depuis PC1 vers S2 :

```
root@poda-1:~# ping 172.16.30.12
PING 172.16.30.12 (172.16.30.12) 56(84) bytes of data.
From 172.16.10.2 icmp_seq=1 Packet filtered
From 172.16.10.2 icmp_seq=2 Packet filtered
From 172.16.10.2 icmp_seq=3 Packet filtered
From 172.16.10.2 icmp_seq=4 Packet filtered
From 172.16.10.2 icmp_seq=5 Packet filtered
From 172.16.10.2 icmp_seq=6 Packet filtered
^C
--- 172.16.30.12 ping statistics ---
6 packets transmitted, 0 received, +6 errors, 100% packet loss, time 5123ms

root@poda-1:~#
```

ping depuis PC1 vers Passerelle Virtuelle (HSRP):

Valide la connectivité avec la passerelle virtuelle et le bon fonctionnement de la redondance HSRP

```
root@poda-1:~# ping 172.16.10.1
PING 172.16.10.1 (172.16.10.1) 56(84) bytes of data.
64 bytes from 172.16.10.1: icmp_seq=1 ttl=255 time=0.696 ms
64 bytes from 172.16.10.1: icmp_seq=2 ttl=255 time=0.650 ms
64 bytes from 172.16.10.1: icmp_seq=3 ttl=255 time=0.705 ms
64 bytes from 172.16.10.1: icmp_seq=4 ttl=255 time=0.811 ms
64 bytes from 172.16.10.1: icmp_seq=5 ttl=255 time=0.616 ms
^C
--- 172.16.10.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4091ms
rtt min/avg/max/mdev = 0.616/0.695/0.811/0.066 ms
root@poda-1:~#
```

ping depuis PC1 vers R1 :

Vérifie la connectivité physique directe entre le commutateur d'accès et l'interface LAN du routeur R1.

```
root@poda-1:~# ping 172.16.10.2
PING 172.16.10.2 (172.16.10.2) 56(84) bytes of data.
64 bytes from 172.16.10.2: icmp_seq=1 ttl=255 time=0.414 ms
64 bytes from 172.16.10.2: icmp_seq=2 ttl=255 time=0.290 ms
64 bytes from 172.16.10.2: icmp_seq=3 ttl=255 time=0.291 ms
64 bytes from 172.16.10.2: icmp_seq=4 ttl=255 time=0.287 ms
64 bytes from 172.16.10.2: icmp_seq=5 ttl=255 time=0.283 ms
64 bytes from 172.16.10.2: icmp_seq=6 ttl=255 time=0.302 ms
^C
--- 172.16.10.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5102ms
rtt min/avg/max/mdev = 0.283/0.311/0.414/0.046 ms
root@poda-1:~#
```

ping depuis PC1 vers R2 :

Vérifie la connectivité physique directe vers le routeur R2, assurant que le chemin de secours est opérationnel.

```
root@poda-1:~# ping 172.16.10.3
PING 172.16.10.3 (172.16.10.3) 56(84) bytes of data.
64 bytes from 172.16.10.3: icmp_seq=1 ttl=255 time=0.929 ms
64 bytes from 172.16.10.3: icmp_seq=2 ttl=255 time=0.998 ms
64 bytes from 172.16.10.3: icmp_seq=3 ttl=255 time=0.892 ms
64 bytes from 172.16.10.3: icmp_seq=4 ttl=255 time=0.690 ms
64 bytes from 172.16.10.3: icmp_seq=5 ttl=255 time=0.644 ms
64 bytes from 172.16.10.3: icmp_seq=6 ttl=255 time=0.983 ms
^C
--- 172.16.10.3 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5044ms
rtt min/avg/max/mdev = 0.644/0.856/0.998/0.138 ms
root@poda-1:~#
```

ping depuis PC1 vers Réseau Dédié R&D 10.10.10.1

Seul le VLAN R&D doit avoir accès à ce réseau

```
root@poda-1:~# ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=254 time=27.3 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=254 time=7.46 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=254 time=8.44 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=254 time=10.4 ms
64 bytes from 10.10.10.1: icmp_seq=5 ttl=254 time=11.9 ms
^C
--- 10.10.10.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 7.459/13.094/27.274/7.255 ms
root@poda-1:~#
```

ping depuis PC1 vers Internet (ISP) 209.165.100.1

Valide que PC1 a accès à Internet via le PAT configuré sur le routeur de bord

```
root@poda-1:~# ping 209.165.100.1
PING 209.165.100.1 (209.165.100.1) 56(84) bytes of data.
64 bytes from 209.165.100.1: icmp_seq=1 ttl=253 time=24.5 ms
64 bytes from 209.165.100.1: icmp_seq=2 ttl=253 time=24.1 ms
64 bytes from 209.165.100.1: icmp_seq=3 ttl=253 time=25.0 ms
64 bytes from 209.165.100.1: icmp_seq=4 ttl=253 time=26.0 ms
64 bytes from 209.165.100.1: icmp_seq=5 ttl=253 time=26.2 ms
^C
--- 209.165.100.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 24.107/25.158/26.170/0.809 ms
root@poda-1:~#
```

ping depuis PC1 vers R1 (Interface de POD-X-R1) :

Ce test prouve que le protocole de routage dynamique **OSPF** est fonctionnel. Le paquet traverse les routeurs R1/R2 pour atteindre l'interface interne du routeur de bord.

```
root@poda-1:~# ping 10.10.13.2
PING 10.10.13.2 (10.10.13.2) 56(84) bytes of data.
64 bytes from 10.10.13.2: icmp_seq=1 ttl=254 time=8.31 ms
64 bytes from 10.10.13.2: icmp_seq=2 ttl=254 time=8.98 ms
64 bytes from 10.10.13.2: icmp_seq=3 ttl=254 time=9.90 ms
64 bytes from 10.10.13.2: icmp_seq=4 ttl=254 time=11.4 ms
64 bytes from 10.10.13.2: icmp_seq=5 ttl=254 time=3.52 ms
^C
--- 10.10.13.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 3.522/8.427/11.433/2.667 ms
root@poda-1:~#
```

ping depuis PC1 vers R2 (Interface de POD-X-R1) :

Valide le second chemin de routage interne vers le routeur de bord, confirmant la stabilité de la table de routage OSPF.

```
root@poda-1:~# ping 10.10.23.2
PING 10.10.23.2 (10.10.23.2) 56(84) bytes of data.
64 bytes from 10.10.23.2: icmp_seq=1 ttl=254 time=9.65 ms
64 bytes from 10.10.23.2: icmp_seq=2 ttl=254 time=9.50 ms
64 bytes from 10.10.23.2: icmp_seq=3 ttl=254 time=9.85 ms
64 bytes from 10.10.23.2: icmp_seq=4 ttl=254 time=9.60 ms
64 bytes from 10.10.23.2: icmp_seq=5 ttl=254 time=9.87 ms
64 bytes from 10.10.23.2: icmp_seq=6 ttl=254 time=10.2 ms
^C
--- 10.10.23.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5007ms
rtt min/avg/max/mdev = 9.496/9.774/10.178/0.224 ms
root@poda-1:~#
```

ping depuis PC1 vers l'interface de sortie du Bord :

Ce test est crucial : il valide la configuration du **NAT/PAT**. Il prouve que l'adresse privée de PC1 est correctement translatée en adresse publique pour sortir du réseau.

```
root@poda-1:~# ping 209.165.200.225
PING 209.165.200.225 (209.165.200.225) 56(84) bytes of data.
64 bytes from 209.165.200.225: icmp_seq=1 ttl=254 time=5.86 ms
64 bytes from 209.165.200.225: icmp_seq=2 ttl=254 time=6.33 ms
64 bytes from 209.165.200.225: icmp_seq=3 ttl=254 time=7.68 ms
64 bytes from 209.165.200.225: icmp_seq=4 ttl=254 time=9.24 ms
64 bytes from 209.165.200.225: icmp_seq=5 ttl=254 time=9.79 ms
64 bytes from 209.165.200.225: icmp_seq=6 ttl=254 time=10.4 ms
64 bytes from 209.165.200.225: icmp_seq=7 ttl=254 time=11.8 ms
^C
--- 209.165.200.225 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6009ms
rtt min/avg/max/mdev = 5.861/8.731/11.814/2.030 ms
root@poda-1:~#
```

ping depuis PC1 vers l'interface Entrée ISP :

Valide la liaison **WAN** (le lien série ou fibre) entre notre entreprise et le fournisseur d'accès. Le "Next-Hop" est atteint avec succès.

```
root@poda-1:~# ping 209.165.200.226
PING 209.165.200.226 (209.165.200.226) 56(84) bytes of data.
64 bytes from 209.165.200.226: icmp_seq=1 ttl=253 time=29.6 ms
64 bytes from 209.165.200.226: icmp_seq=2 ttl=253 time=30.4 ms
64 bytes from 209.165.200.226: icmp_seq=3 ttl=253 time=30.6 ms
64 bytes from 209.165.200.226: icmp_seq=4 ttl=253 time=31.1 ms
64 bytes from 209.165.200.226: icmp_seq=5 ttl=253 time=32.0 ms
64 bytes from 209.165.200.226: icmp_seq=6 ttl=253 time=32.7 ms
64 bytes from 209.165.200.226: icmp_seq=7 ttl=253 time=23.6 ms
64 bytes from 209.165.200.226: icmp_seq=8 ttl=253 time=24.4 ms
^C
--- 209.165.200.226 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7011ms
rtt min/avg/max/mdev = 23.573/29.300/32.733/3.214 ms
root@poda-1:~#
```

ping depuis PC2 vers PC1

```
root@poda-2:~# ping 172.16.10.10
PING 172.16.10.10 (172.16.10.10) 56(84) bytes of data.
64 bytes from 172.16.10.10: icmp_seq=2 ttl=63 time=0.804 ms
64 bytes from 172.16.10.10: icmp_seq=3 ttl=63 time=0.787 ms
64 bytes from 172.16.10.10: icmp_seq=4 ttl=63 time=0.528 ms
64 bytes from 172.16.10.10: icmp_seq=5 ttl=63 time=0.541 ms
64 bytes from 172.16.10.10: icmp_seq=6 ttl=63 time=0.549 ms
64 bytes from 172.16.10.10: icmp_seq=7 ttl=63 time=0.517 ms
64 bytes from 172.16.10.10: icmp_seq=8 ttl=63 time=0.497 ms
^C
--- 172.16.10.10 ping statistics ---
8 packets transmitted, 7 received, 12.5% packet loss, time 7156ms
rtt min/avg/max/mdev = 0.497/0.603/0.804/0.122 ms
root@poda-2:~#
```

ping depuis PC2 vers PC3

```
root@poda-2:~# ping 172.16.30.10
PING 172.16.30.10 (172.16.30.10) 56(84) bytes of data.
64 bytes from 172.16.30.10: icmp_seq=1 ttl=63 time=0.772 ms
64 bytes from 172.16.30.10: icmp_seq=2 ttl=63 time=0.773 ms
64 bytes from 172.16.30.10: icmp_seq=3 ttl=63 time=0.746 ms
64 bytes from 172.16.30.10: icmp_seq=4 ttl=63 time=0.783 ms
64 bytes from 172.16.30.10: icmp_seq=5 ttl=63 time=0.773 ms
64 bytes from 172.16.30.10: icmp_seq=6 ttl=63 time=0.925 ms
64 bytes from 172.16.30.10: icmp_seq=7 ttl=63 time=0.778 ms
^C
--- 172.16.30.10 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6122ms
rtt min/avg/max/mdev = 0.746/0.792/0.925/0.055 ms
root@poda-2:~#
```

ping depuis PC2 vers S1

```
root@poda-2:~# ping 172.16.30.11
PING 172.16.30.11 (172.16.30.11) 56(84) bytes of data.
64 bytes from 172.16.30.11: icmp_seq=1 ttl=254 time=0.806 ms
64 bytes from 172.16.30.11: icmp_seq=2 ttl=254 time=0.799 ms
64 bytes from 172.16.30.11: icmp_seq=3 ttl=254 time=0.807 ms
64 bytes from 172.16.30.11: icmp_seq=4 ttl=254 time=0.834 ms
64 bytes from 172.16.30.11: icmp_seq=5 ttl=254 time=0.781 ms
64 bytes from 172.16.30.11: icmp_seq=6 ttl=254 time=0.809 ms
64 bytes from 172.16.30.11: icmp_seq=7 ttl=254 time=0.811 ms
^C
--- 172.16.30.11 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6145ms
rtt min/avg/max/mdev = 0.781/0.806/0.834/0.014 ms
root@poda-2:~#
```

ping depuis PC2 vers S2

```
root@poda-2:~# ping 172.16.30.12
PING 172.16.30.12 (172.16.30.12) 56(84) bytes of data.
From 172.16.20.2 icmp_seq=1 Packet filtered
From 172.16.20.2 icmp_seq=2 Packet filtered
From 172.16.20.2 icmp_seq=3 Packet filtered
From 172.16.20.2 icmp_seq=4 Packet filtered
From 172.16.20.2 icmp_seq=5 Packet filtered
From 172.16.20.2 icmp_seq=6 Packet filtered
From 172.16.20.2 icmp_seq=7 Packet filtered
^C
--- 172.16.30.12 ping statistics ---
7 packets transmitted, 0 received, +7 errors, 100% packet loss, time 6124ms

root@poda-2:~#
```

ping depuis PC2 vers Passerelle Virtuelle :

Valide que PC2 (VLAN 20) communique avec l'IP virtuelle **HSRP**. C'est le premier saut pour sortir du VLAN.

```
root@poda-2:~# ping 172.16.20.1
PING 172.16.20.1 (172.16.20.1) 56(84) bytes of data.
64 bytes from 172.16.20.1: icmp_seq=1 ttl=255 time=0.683 ms
64 bytes from 172.16.20.1: icmp_seq=2 ttl=255 time=0.672 ms
64 bytes from 172.16.20.1: icmp_seq=3 ttl=255 time=0.689 ms
64 bytes from 172.16.20.1: icmp_seq=4 ttl=255 time=0.754 ms
64 bytes from 172.16.20.1: icmp_seq=5 ttl=255 time=0.728 ms
64 bytes from 172.16.20.1: icmp_seq=6 ttl=255 time=0.768 ms
64 bytes from 172.16.20.1: icmp_seq=7 ttl=255 time=0.745 ms
^C
--- 172.16.20.1 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6139ms
rtt min/avg/max/mdev = 0.672/0.719/0.768/0.035 ms
root@poda-2:~#
```

ping depuis PC2 vers R1

Vérifie la connectivité physique directe vers l'interface LAN du routeur R1 pour le VLAN 20.

```
root@poda-2:~# ping 172.16.20.2
PING 172.16.20.2 (172.16.20.2) 56(84) bytes of data.
64 bytes from 172.16.20.2: icmp_seq=1 ttl=255 time=1.21 ms
64 bytes from 172.16.20.2: icmp_seq=2 ttl=255 time=0.346 ms
64 bytes from 172.16.20.2: icmp_seq=3 ttl=255 time=0.357 ms
64 bytes from 172.16.20.2: icmp_seq=4 ttl=255 time=0.378 ms
64 bytes from 172.16.20.2: icmp_seq=5 ttl=255 time=0.338 ms
64 bytes from 172.16.20.2: icmp_seq=6 ttl=255 time=0.384 ms
64 bytes from 172.16.20.2: icmp_seq=7 ttl=255 time=0.323 ms
^C
--- 172.16.20.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6117ms
rtt min/avg/max/mdev = 0.323/0.476/1.210/0.300 ms
root@poda-2:~#
```

ping depuis PC2 vers R2

Vérifie la connectivité vers le routeur R2, confirmant que le chemin de redondance est prêt.

```
root@poda-2:~# ping 172.16.20.3
PING 172.16.20.3 (172.16.20.3) 56(84) bytes of data.
64 bytes from 172.16.20.3: icmp_seq=1 ttl=255 time=0.871 ms
64 bytes from 172.16.20.3: icmp_seq=2 ttl=255 time=0.620 ms
64 bytes from 172.16.20.3: icmp_seq=3 ttl=255 time=0.692 ms
64 bytes from 172.16.20.3: icmp_seq=4 ttl=255 time=0.655 ms
64 bytes from 172.16.20.3: icmp_seq=5 ttl=255 time=0.845 ms
64 bytes from 172.16.20.3: icmp_seq=6 ttl=255 time=0.985 ms
64 bytes from 172.16.20.3: icmp_seq=7 ttl=255 time=0.645 ms
64 bytes from 172.16.20.3: icmp_seq=8 ttl=255 time=0.845 ms
64 bytes from 172.16.20.3: icmp_seq=9 ttl=255 time=0.662 ms
^C
--- 172.16.20.3 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8142ms
rtt min/avg/max/mdev = 0.620/0.757/0.985/0.122 ms
root@poda-2:~#
```

ping depuis PC2 vers l'interface ISP :

Prouve que le VLAN 20 peut sortir du réseau. Si tu as configuré un **NAT Statique** pour ce PC (serveur), ce test confirme que la translation fonctionne.

```
root@poda-2:~# ping 209.165.200.226
PING 209.165.200.226 (209.165.200.226) 56(84) bytes of data.
64 bytes from 209.165.200.226: icmp_seq=1 ttl=253 time=32.8 ms
64 bytes from 209.165.200.226: icmp_seq=2 ttl=253 time=23.3 ms
64 bytes from 209.165.200.226: icmp_seq=3 ttl=253 time=32.4 ms
64 bytes from 209.165.200.226: icmp_seq=4 ttl=253 time=22.7 ms
64 bytes from 209.165.200.226: icmp_seq=5 ttl=253 time=12.7 ms
64 bytes from 209.165.200.226: icmp_seq=6 ttl=253 time=13.1 ms
^C
--- 209.165.200.226 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5009ms
rtt min/avg/max/mdev = 12.702/22.851/32.844/8.056 ms
root@poda-2:~#
```

ping depuis PC2 vers Internet (ISP):

Confirme que le Staff (VLAN 20) a un accès complet à l'extérieur via le routage OSPF et le NAT.

```
root@poda-2:~# ping 209.165.100.1
PING 209.165.100.1 (209.165.100.1) 56(84) bytes of data.
64 bytes from 209.165.100.1: icmp_seq=1 ttl=253 time=21.5 ms
64 bytes from 209.165.100.1: icmp_seq=2 ttl=253 time=22.1 ms
64 bytes from 209.165.100.1: icmp_seq=3 ttl=253 time=24.2 ms
64 bytes from 209.165.100.1: icmp_seq=4 ttl=253 time=26.0 ms
64 bytes from 209.165.100.1: icmp_seq=5 ttl=253 time=27.0 ms
64 bytes from 209.165.100.1: icmp_seq=6 ttl=253 time=28.0 ms
64 bytes from 209.165.100.1: icmp_seq=7 ttl=253 time=28.2 ms
^C
--- 209.165.100.1 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6006ms
rtt min/avg/max/mdev = 21.542/25.293/28.187/2.520 ms
root@poda-2:~#
```

ping depuis PC2 vers Réseau R&D 10.10.10.1

Ce réseau est strictement réservé au VLAN R&D ; le Staff est bloqué.

```
root@poda-2:~# ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
From 172.16.20.2 icmp_seq=1 Packet filtered
From 172.16.20.2 icmp_seq=2 Packet filtered
From 172.16.20.2 icmp_seq=3 Packet filtered
From 172.16.20.2 icmp_seq=4 Packet filtered
From 172.16.20.2 icmp_seq=5 Packet filtered
From 172.16.20.2 icmp_seq=6 Packet filtered
^C
--- 10.10.10.1 ping statistics ---
6 packets transmitted, 0 received, +6 errors, 100% packet loss, time 5103ms

root@poda-2:~#
```

ping depuis PC2 vers Lien R2 (Bord) 10.10.23.2

Valide que le routage OSPF est fonctionnel pour acheminer le trafic du serveur

```
root@poda-2:~# ping 10.10.23.2
PING 10.10.23.2 (10.10.23.2) 56(84) bytes of data.
64 bytes from 10.10.23.2: icmp_seq=1 ttl=254 time=7.71 ms
64 bytes from 10.10.23.2: icmp_seq=2 ttl=254 time=9.00 ms
64 bytes from 10.10.23.2: icmp_seq=3 ttl=254 time=10.3 ms
64 bytes from 10.10.23.2: icmp_seq=4 ttl=254 time=12.1 ms
64 bytes from 10.10.23.2: icmp_seq=5 ttl=254 time=3.20 ms
64 bytes from 10.10.23.2: icmp_seq=6 ttl=254 time=5.13 ms
^C
--- 10.10.23.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5001ms
rtt min/avg/max/mdev = 3.201/7.899/12.078/3.004 ms
root@poda-2:~#
```

ping depuis PC3 vers PC1

Cela prouve que l'admin peut surveiller les postes clients

```
root@poda-3:~# ping 172.16.10.10
PING 172.16.10.10 (172.16.10.10) 56(84) bytes of data.
64 bytes from 172.16.10.10: icmp_seq=1 ttl=63 time=0.753 ms
64 bytes from 172.16.10.10: icmp_seq=2 ttl=63 time=0.489 ms
64 bytes from 172.16.10.10: icmp_seq=3 ttl=63 time=0.525 ms
64 bytes from 172.16.10.10: icmp_seq=4 ttl=63 time=0.503 ms
64 bytes from 172.16.10.10: icmp_seq=5 ttl=63 time=0.513 ms
64 bytes from 172.16.10.10: icmp_seq=6 ttl=63 time=0.500 ms
^C
--- 172.16.10.10 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5098ms
rtt min/avg/max/mdev = 0.489/0.547/0.753/0.092 ms
root@poda-3:~#
```

ping depuis PC3 vers PC2

Permet à l'administrateur de vérifier l'état du serveur Staff.

```
root@poda-3:~# ping 172.16.20.10
PING 172.16.20.10 (172.16.20.10) 56(84) bytes of data.
64 bytes from 172.16.20.10: icmp_seq=1 ttl=63 time=0.522 ms
64 bytes from 172.16.20.10: icmp_seq=2 ttl=63 time=0.522 ms
64 bytes from 172.16.20.10: icmp_seq=3 ttl=63 time=0.772 ms
64 bytes from 172.16.20.10: icmp_seq=4 ttl=63 time=0.787 ms
64 bytes from 172.16.20.10: icmp_seq=5 ttl=63 time=0.769 ms
64 bytes from 172.16.20.10: icmp_seq=6 ttl=63 time=0.795 ms
^C
--- 172.16.20.10 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5125ms
rtt min/avg/max/mdev = 0.522/0.694/0.795/0.122 ms
root@poda-3:~#
```

ping depuis PC3 vers s1

Prouve que l'administrateur peut joindre l'interface de gestion du commutateur S1.

```
root@poda-3:~# ping 172.16.30.11
PING 172.16.30.11 (172.16.30.11) 56(84) bytes of data.
64 bytes from 172.16.30.11: icmp_seq=2 ttl=255 time=0.598 ms
64 bytes from 172.16.30.11: icmp_seq=3 ttl=255 time=0.587 ms
64 bytes from 172.16.30.11: icmp_seq=4 ttl=255 time=0.615 ms
64 bytes from 172.16.30.11: icmp_seq=5 ttl=255 time=0.615 ms
64 bytes from 172.16.30.11: icmp_seq=6 ttl=255 time=0.582 ms
64 bytes from 172.16.30.11: icmp_seq=7 ttl=255 time=0.584 ms
64 bytes from 172.16.30.11: icmp_seq=8 ttl=255 time=0.594 ms
^C
--- 172.16.30.11 ping statistics ---
8 packets transmitted, 7 received, 12.5% packet loss, time 7145ms
rtt min/avg/max/mdev = 0.582/0.596/0.615/0.012 ms
root@poda-3:~#
```

ping depuis PC3 vers s2

Confirme la connectivité avec le second commutateur du cœur de réseau

```
root@poda-3:~# ping 172.16.30.12
PING 172.16.30.12 (172.16.30.12) 56(84) bytes of data.
64 bytes from 172.16.30.12: icmp_seq=2 ttl=255 time=0.587 ms
64 bytes from 172.16.30.12: icmp_seq=3 ttl=255 time=0.574 ms
64 bytes from 172.16.30.12: icmp_seq=4 ttl=255 time=0.591 ms
64 bytes from 172.16.30.12: icmp_seq=5 ttl=255 time=0.581 ms
64 bytes from 172.16.30.12: icmp_seq=6 ttl=255 time=0.581 ms
^C
--- 172.16.30.12 ping statistics ---
6 packets transmitted, 5 received, 16.6667% packet loss, time 5120ms
rtt min/avg/max/mdev = 0.574/0.582/0.591/0.005 ms
root@poda-3:~#
```

ping depuis PC3 vers la passerelle virtuelle

Valide que le VLAN 30 communique avec sa passerelle de redondance

```
root@poda-3:~# ping 172.16.30.1
PING 172.16.30.1 (172.16.30.1) 56(84) bytes of data.
64 bytes from 172.16.30.1: icmp_seq=1 ttl=255 time=0.667 ms
64 bytes from 172.16.30.1: icmp_seq=2 ttl=255 time=0.725 ms
64 bytes from 172.16.30.1: icmp_seq=3 ttl=255 time=0.655 ms
64 bytes from 172.16.30.1: icmp_seq=4 ttl=255 time=0.704 ms
64 bytes from 172.16.30.1: icmp_seq=5 ttl=255 time=0.731 ms
64 bytes from 172.16.30.1: icmp_seq=6 ttl=255 time=0.690 ms
64 bytes from 172.16.30.1: icmp_seq=7 ttl=255 time=0.740 ms
^C
--- 172.16.30.1 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6134ms
rtt min/avg/max/mdev = 0.655/0.701/0.740/0.030 ms
root@poda-3:~#
```

ping depuis PC3 vers R1

Vérifie que l'admin peut joindre le routeur R1 par son interface locale

```
root@poda-3:~# ping 172.16.30.2
PING 172.16.30.2 (172.16.30.2) 56(84) bytes of data.
64 bytes from 172.16.30.2: icmp_seq=1 ttl=255 time=0.448 ms
64 bytes from 172.16.30.2: icmp_seq=2 ttl=255 time=0.329 ms
64 bytes from 172.16.30.2: icmp_seq=3 ttl=255 time=0.327 ms
64 bytes from 172.16.30.2: icmp_seq=4 ttl=255 time=0.334 ms
64 bytes from 172.16.30.2: icmp_seq=5 ttl=255 time=0.338 ms
64 bytes from 172.16.30.2: icmp_seq=6 ttl=255 time=0.312 ms
64 bytes from 172.16.30.2: icmp_seq=7 ttl=255 time=0.311 ms
^C
--- 172.16.30.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6132ms
rtt min/avg/max/mdev = 0.311/0.342/0.448/0.044 ms
root@poda-3:~#
```

ping depuis PC3 vers R2

Vérifie la connectivité vers le routeur R2 pour l'administration

```
root@poda-3:~# ping 172.16.30.3
PING 172.16.30.3 (172.16.30.3) 56(84) bytes of data.
64 bytes from 172.16.30.3: icmp_seq=2 ttl=255 time=0.799 ms
64 bytes from 172.16.30.3: icmp_seq=3 ttl=255 time=0.612 ms
64 bytes from 172.16.30.3: icmp_seq=4 ttl=255 time=0.815 ms
64 bytes from 172.16.30.3: icmp_seq=5 ttl=255 time=0.605 ms
64 bytes from 172.16.30.3: icmp_seq=6 ttl=255 time=0.689 ms
64 bytes from 172.16.30.3: icmp_seq=7 ttl=255 time=0.964 ms
64 bytes from 172.16.30.3: icmp_seq=8 ttl=255 time=0.620 ms
64 bytes from 172.16.30.3: icmp_seq=9 ttl=255 time=0.973 ms
^C
--- 172.16.30.3 ping statistics ---
9 packets transmitted, 8 received, 11.1111% packet loss, time 8166ms
rtt min/avg/max/mdev = 0.605/0.759/0.973/0.142 ms
root@poda-3:~#
```

ping depuis PC3 vers l'interface ISP

Prouve que le trafic du VLAN 30 est correctement translaté par le NAT dynamique du routeur de bord

```
root@poda-3:~# ping 209.165.200.226
PING 209.165.200.226 (209.165.200.226) 56(84) bytes of data.
64 bytes from 209.165.200.226: icmp_seq=1 ttl=253 time=24.9 ms
64 bytes from 209.165.200.226: icmp_seq=2 ttl=253 time=25.2 ms
64 bytes from 209.165.200.226: icmp_seq=3 ttl=253 time=26.1 ms
64 bytes from 209.165.200.226: icmp_seq=4 ttl=253 time=27.4 ms
64 bytes from 209.165.200.226: icmp_seq=5 ttl=253 time=28.3 ms
64 bytes from 209.165.200.226: icmp_seq=6 ttl=253 time=19.2 ms
64 bytes from 209.165.200.226: icmp_seq=7 ttl=253 time=29.5 ms
^C
--- 209.165.200.226 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6007ms
rtt min/avg/max/mdev = 19.192/25.812/29.505/3.107 ms
root@poda-3:~#
```

ping depuis PC3 vers Internet 209.165.100.1

Connectivité totale de bout en bout pour le poste administrateur

```
root@poda-3:~# ping 209.165.100.1
PING 209.165.100.1 (209.165.100.1) 56(84) bytes of data.
64 bytes from 209.165.100.1: icmp_seq=1 ttl=253 time=16.7 ms
64 bytes from 209.165.100.1: icmp_seq=2 ttl=253 time=17.0 ms
64 bytes from 209.165.100.1: icmp_seq=3 ttl=253 time=18.0 ms
64 bytes from 209.165.100.1: icmp_seq=4 ttl=253 time=19.6 ms
64 bytes from 209.165.100.1: icmp_seq=5 ttl=253 time=30.5 ms
64 bytes from 209.165.100.1: icmp_seq=6 ttl=253 time=30.2 ms
64 bytes from 209.165.100.1: icmp_seq=7 ttl=253 time=31.5 ms
^C
--- 209.165.100.1 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6008ms
rtt min/avg/max/mdev = 16.657/23.346/31.528/6.475 ms
root@poda-3:~#
```

ping depuis PC3 vers Bord (Interface Interne) 10.10.13.2

Prouve que le routage **OSPF** permet à l'admin de traverser R1 pour atteindre le routeur de bord

```
root@poda-3:~# ping 10.10.13.2
PING 10.10.13.2 (10.10.13.2) 56(84) bytes of data.
64 bytes from 10.10.13.2: icmp_seq=1 ttl=254 time=9.76 ms
64 bytes from 10.10.13.2: icmp_seq=2 ttl=254 time=9.22 ms
64 bytes from 10.10.13.2: icmp_seq=3 ttl=254 time=8.79 ms
64 bytes from 10.10.13.2: icmp_seq=4 ttl=254 time=8.60 ms
64 bytes from 10.10.13.2: icmp_seq=5 ttl=254 time=8.75 ms
64 bytes from 10.10.13.2: icmp_seq=6 ttl=254 time=8.28 ms
^C
--- 10.10.13.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5009ms
rtt min/avg/max/mdev = 8.283/8.898/9.761/0.474 ms
root@poda-3:~#
```

ping depuis PC3 vers Bord (Interface WAN) 209.165.200.225

prouve que l'admin peut atteindre l'interface publique de son propre routeur de bord

```
root@poda-3:~# ping 209.165.200.225
PING 209.165.200.225 (209.165.200.225) 56(84) bytes of data.
64 bytes from 209.165.200.225: icmp_seq=1 ttl=254 time=12.6 ms
64 bytes from 209.165.200.225: icmp_seq=2 ttl=254 time=2.93 ms
64 bytes from 209.165.200.225: icmp_seq=3 ttl=254 time=3.27 ms
64 bytes from 209.165.200.225: icmp_seq=4 ttl=254 time=3.11 ms
64 bytes from 209.165.200.225: icmp_seq=5 ttl=254 time=3.71 ms
64 bytes from 209.165.200.225: icmp_seq=6 ttl=254 time=3.46 ms
64 bytes from 209.165.200.225: icmp_seq=7 ttl=254 time=3.84 ms
^C
--- 209.165.200.225 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6009ms
rtt min/avg/max/mdev = 2.934/4.701/12.585/3.231 ms
root@poda-3:~#
```

ping depuis PC3 vers R2 (Interface OSPF) 10.10.23.1

Vérifie que l'admin peut joindre l'interface de "sortie" du routeur R2 vers le bord

```
root@poda-3:~# ping 10.10.23.1
PING 10.10.23.1 (10.10.23.1) 56(84) bytes of data.
64 bytes from 10.10.23.1: icmp_seq=1 ttl=255 time=0.713 ms
64 bytes from 10.10.23.1: icmp_seq=2 ttl=255 time=0.983 ms
64 bytes from 10.10.23.1: icmp_seq=3 ttl=255 time=0.982 ms
64 bytes from 10.10.23.1: icmp_seq=4 ttl=255 time=0.839 ms
64 bytes from 10.10.23.1: icmp_seq=5 ttl=255 time=0.808 ms
^C
--- 10.10.23.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4010ms
rtt min/avg/max/mdev = 0.713/0.865/0.983/0.104 ms
root@poda-3:~#
```

ping depuis PC3 vers Bord (Interface via R2) 10.10.23.2

Prouve que le routage fonctionne aussi via le chemin de droite (R2)

```
root@poda-3:~# ping 10.10.23.2
PING 10.10.23.2 (10.10.23.2) 56(84) bytes of data.
64 bytes from 10.10.23.2: icmp_seq=1 ttl=254 time=4.79 ms
64 bytes from 10.10.23.2: icmp_seq=2 ttl=254 time=6.10 ms
64 bytes from 10.10.23.2: icmp_seq=3 ttl=254 time=7.65 ms
64 bytes from 10.10.23.2: icmp_seq=4 ttl=254 time=9.04 ms
64 bytes from 10.10.23.2: icmp_seq=5 ttl=254 time=9.35 ms
^C
--- 10.10.23.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 4.790/7.384/9.350/1.735 ms
root@poda-3:~#
```

ping depuis PC3 vers Réseau Dédié R&D 10.10.10.1

Echec , ce réseau est réservé exclusivement au VLAN R&D

```
root@poda-3:~# ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
^C
--- 10.10.10.1 ping statistics ---
16 packets transmitted, 0 received, 100% packet loss, time 15368ms

root@poda-3:~#
```

Test SSH (Pour prouver l'accès admin) depuis PC3 vers S1

Échec de la connexion SSH dû à une incompatibilité de version du protocole. L'équipement distant (172.16.30.11) utilise exclusivement **SSH v1**, une version obsolète, tandis que le client local impose **SSH v2** pour des raisons de sécurité.

```
root@poda-3:~# ssh admin@172.16.30.11
Protocol major versions differ: 2 vs. 1
banner exchange: Connection to 172.16.30.11 port 22: could not read protocol version
root@poda-3:~#
```

show etherchannel summary sur S1 :

Cette commande valide la création du lien logique entre les commutateurs. L'état **(SU)** confirme que l'agrégation est active au niveau 2 et opérationnelle, permettant de doubler la bande passante entre S1 et S2 tout en offrant une redondance physique

```
S1#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Fa0/1(P)    Fa0/2(P)

S1#
```

show etherchannel summary sur S2 :

```
S2#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Fa0/1(P)    Fa0/2(P)

S2#
```

show interfaces trunk sur S1 :

```
S1#sh interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Fa0/5       on               802.1q         trunking      99
Po1         on               802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/5       1-4094
Po1         1-4094

Port        Vlans allowed and active in management domain
Fa0/5       1,10,20,30,40,90,99
Po1         1,10,20,30,40,90,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       1,10,20,30,40,90,99
Po1         1,10,20,30,40,90,99
S1#
```

show interfaces trunk sur S2 :

```
S2#sh interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Fa0/5       on               802.1q         trunking      99
Po1         on               802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/5       1-4094
Po1         1-4094

Port        Vlans allowed and active in management domain
Fa0/5       1,10,20,30,40,90,99
Po1         1,10,20,30,40,90,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       1,10,20,30,40,90,99
Po1         1,10,20,30,40,90,99
S2#
```

show vlan brief sur S1 :

```
S1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   RD                               active    Fa0/6
20   STAFF                            active
30   MGNT                             active
40   OSPF_PC0                         active
90   TRASH                            active    Fa0/3, Fa0/4, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gi0/1, Gi0/2
99   NATIF                            active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
S1#
```

show vlan brief sur S2 :

```
S2#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   RD                               active
20   STAFF                            active    Fa0/24
30   MGNT                             active    Fa0/18
40   OSPF_PC0                         active
90   TRASH                            active    Fa0/3, Fa0/4, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Gi0/1
                                                Gi0/2
99   NATIF                            active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
S2#
```

sh ip int br sur S1 :

```
S1#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
Vlan1                  unassigned      YES NVRAM  administratively down down
Vlan30                 172.16.30.11    YES NVRAM  up                    up
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  up                    up
FastEthernet0/3        unassigned      YES unset  administratively down down
FastEthernet0/4        unassigned      YES unset  administratively down down
FastEthernet0/5        unassigned      YES unset  up                    up
FastEthernet0/6        unassigned      YES unset  up                    up
FastEthernet0/7        unassigned      YES unset  administratively down down
FastEthernet0/8        unassigned      YES unset  administratively down down
FastEthernet0/9        unassigned      YES unset  administratively down down
FastEthernet0/10       unassigned      YES unset  administratively down down
FastEthernet0/11       unassigned      YES unset  administratively down down
FastEthernet0/12       unassigned      YES unset  administratively down down
FastEthernet0/13       unassigned      YES unset  administratively down down
FastEthernet0/14       unassigned      YES unset  administratively down down
FastEthernet0/15       unassigned      YES unset  administratively down down
FastEthernet0/16       unassigned      YES unset  administratively down down
FastEthernet0/17       unassigned      YES unset  administratively down down
FastEthernet0/18       unassigned      YES unset  administratively down down
FastEthernet0/19       unassigned      YES unset  administratively down down
FastEthernet0/20       unassigned      YES unset  administratively down down
FastEthernet0/21       unassigned      YES unset  administratively down down
FastEthernet0/22       unassigned      YES unset  administratively down down
FastEthernet0/23       unassigned      YES unset  administratively down down
FastEthernet0/24       unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Port-channel1          unassigned      YES unset  up                    up
S1#
```

sh ip int br sur S2:

```
S2#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
Vlan1                  unassigned      YES NVRAM  administratively down down
Vlan30                 172.16.30.12    YES NVRAM  up                    up
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  up                    up
FastEthernet0/3        unassigned      YES unset  administratively down down
FastEthernet0/4        unassigned      YES unset  administratively down down
FastEthernet0/5        unassigned      YES unset  up                    up
FastEthernet0/6        unassigned      YES unset  administratively down down
FastEthernet0/7        unassigned      YES unset  administratively down down
FastEthernet0/8        unassigned      YES unset  administratively down down
FastEthernet0/9        unassigned      YES unset  administratively down down
FastEthernet0/10       unassigned      YES unset  administratively down down
FastEthernet0/11       unassigned      YES unset  administratively down down
FastEthernet0/12       unassigned      YES unset  administratively down down
FastEthernet0/13       unassigned      YES unset  administratively down down
FastEthernet0/14       unassigned      YES unset  administratively down down
FastEthernet0/15       unassigned      YES unset  administratively down down
FastEthernet0/16       unassigned      YES unset  administratively down down
FastEthernet0/17       unassigned      YES unset  administratively down down
FastEthernet0/18       unassigned      YES unset  up                    up
FastEthernet0/19       unassigned      YES unset  administratively down down
FastEthernet0/20       unassigned      YES unset  administratively down down
FastEthernet0/21       unassigned      YES unset  administratively down down
FastEthernet0/22       unassigned      YES unset  administratively down down
FastEthernet0/23       unassigned      YES unset  administratively down down
FastEthernet0/24       unassigned      YES unset  up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Port-channel1          unassigned      YES unset  up                    up
S2#
```

sh ip inter br sur R1

```
R1#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES NVRAM  up                    up
Gi0/0/0.10             172.16.10.2     YES NVRAM  up                    up
Gi0/0/0.20             172.16.20.2     YES NVRAM  up                    up
Gi0/0/0.30             172.16.30.2     YES NVRAM  up                    up
GigabitEthernet0/0/1   10.10.13.1       YES NVRAM  up                    up
GigabitEthernet0/0/2   unassigned      YES NVRAM  administratively down down
GigabitEthernet0       unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES NVRAM  administratively down down
R1#
```

sh ip route sur R1

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.10.13.2 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.10.13.2, 00:57:40, GigabitEthernet0/0/1
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.10.13.0/30 is directly connected, GigabitEthernet0/0/1
L        10.10.13.1/32 is directly connected, GigabitEthernet0/0/1
O        10.10.23.0/30
           [110/2] via 172.16.30.3, 00:43:17, GigabitEthernet0/0/0.30
           [110/2] via 172.16.20.3, 00:43:17, GigabitEthernet0/0/0.20
           [110/2] via 172.16.10.3, 00:43:27, GigabitEthernet0/0/0.10
      172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C        172.16.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L        172.16.10.2/32 is directly connected, GigabitEthernet0/0/0.10
C        172.16.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L        172.16.20.2/32 is directly connected, GigabitEthernet0/0/0.20
C        172.16.30.0/24 is directly connected, GigabitEthernet0/0/0.30
L        172.16.30.2/32 is directly connected, GigabitEthernet0/0/0.30
R1#
```

sh standby brief sur R1

```
R1#sh standby br
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Gi0/0/0.10  10   110 P Active  local           172.16.10.3     172.16.10.1
Gi0/0/0.20  20   110 P Active  local           172.16.20.3     172.16.20.1
Gi0/0/0.30  30   110 P Active  local           172.16.30.3     172.16.30.1
R1#
```

sh ip inter br sur R2

```
R2#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES NVRAM  up                    up
Gi0/0/0.10             172.16.10.3     YES NVRAM  up                    up
Gi0/0/0.20             172.16.20.3     YES NVRAM  up                    up
Gi0/0/0.30             172.16.30.3     YES NVRAM  up                    up
GigabitEthernet0/0/1   10.10.23.1      YES NVRAM  up                    up
GigabitEthernet0/0/2   unassigned      YES NVRAM  administratively down down
GigabitEthernet0       unassigned      YES NVRAM  administratively down down
R2#
```

sh ip route sur R2

```
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.10.23.2 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.10.23.2, 00:58:13, GigabitEthernet0/0/1
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.10.13.0/30
           [110/2] via 172.16.30.2, 00:45:33, GigabitEthernet0/0/0.30
           [110/2] via 172.16.20.2, 00:45:33, GigabitEthernet0/0/0.20
           [110/2] via 172.16.10.2, 00:45:38, GigabitEthernet0/0/0.10
C        10.10.23.0/30 is directly connected, GigabitEthernet0/0/1
L        10.10.23.1/32 is directly connected, GigabitEthernet0/0/1
      172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C        172.16.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L        172.16.10.3/32 is directly connected, GigabitEthernet0/0/0.10
C        172.16.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L        172.16.20.3/32 is directly connected, GigabitEthernet0/0/0.20
C        172.16.30.0/24 is directly connected, GigabitEthernet0/0/0.30
L        172.16.30.3/32 is directly connected, GigabitEthernet0/0/0.30
R2#
```

sh ip dhcp binding sur R2

c'est R3 qui gère l'attribution des adresses IP pour tout mon réseau, et non R1 ou R2. Alors ce résultat est normal

```
R2#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address       Client-ID/              Lease expiration        Type       State      Interface
                 Hardware address/
                 User name
R2#
```

sh standby brief sur R2

```
R2#sh standby br
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Gi0/0/0.10  10   100 P Standby 172.16.10.2     local           172.16.10.1
Gi0/0/0.20  20   100 P Standby 172.16.20.2     local           172.16.20.1
Gi0/0/0.30  30   100 P Standby 172.16.30.2     local           172.16.30.1
R2#
```

sh ip dhcp binding sur POD-X-R1

```
POD-X-R1#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
POD-X-R1#
```

sh ip inter br sur POD-X-R1

```
POD-X-R1#sh ip int br
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.10.13.2      YES manual up                    up
Serial0/0                  unassigned      YES unset  administratively down down
FastEthernet0/1            10.10.23.2      YES manual up                    up
Serial0/1                  unassigned      YES unset  administratively down down
FastEthernet1/0            209.165.200.225 YES manual up                    up
FastEthernet2/0            unassigned      YES unset  administratively down down
FastEthernet3/0            unassigned      YES unset  administratively down down
FastEthernet4/0            unassigned      YES unset  administratively down down
NVI0                       unassigned      NO  unset  up                    up
POD-X-R1#
```

sh ip interface sur POD-X-R1 :

```
POD-X-R1#sh ip int
FastEthernet0/0 is up, line protocol is up
  Internet address is 10.10.13.2/30
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Multicast reserved groups joined: 224.0.0.5 224.0.0.6
  Outgoing access list is not set
  Inbound  access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP CEF switching is enabled
  IP CEF Feature Fast switching turbo vector
  IP multicast fast switching is enabled
  IP multicast distributed fast switching is disabled
  IP route-cache flags are Fast, CEF
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Policy routing is disabled
  Network address translation is enabled, interface in domain inside
  BGP Policy Mapping is disabled
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled
Serial0/0 is administratively down, line protocol is down
  Internet protocol processing disabled
FastEthernet0/1 is up, line protocol is up
  Internet address is 10.10.23.2/30
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Multicast reserved groups joined: 224.0.0.5 224.0.0.6
 --More--
```

```
  IP multicast distributed fast switching is disabled
  IP route-cache flags are Fast, CEF
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Policy routing is disabled
  Network address translation is enabled, interface in domain inside
  BGP Policy Mapping is disabled
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled
Serial0/0 is administratively down, line protocol is down
  Internet protocol processing disabled
FastEthernet0/1 is up, line protocol is up
  Internet address is 10.10.23.2/30
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Multicast reserved groups joined: 224.0.0.5 224.0.0.6
  Outgoing access list is not set
  Inbound  access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP CEF switching is enabled
  IP CEF Feature Fast switching turbo vector
  IP multicast fast switching is enabled
  IP multicast distributed fast switching is disabled
  IP route-cache flags are Fast, CEF
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Policy routing is disabled
```

sh ip route sur POD-X-R1 :

```
POD-X-R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     172.16.0.0/24 is subnetted, 3 subnets
O       172.16.30.0 [110/11] via 10.10.23.1, 00:49:01, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:49:01, FastEthernet0/0
O       172.16.20.0 [110/11] via 10.10.23.1, 00:49:01, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:49:01, FastEthernet0/0
O       172.16.10.0 [110/11] via 10.10.23.1, 00:49:01, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:49:01, FastEthernet0/0
     209.165.200.0/27 is subnetted, 1 subnets
C       209.165.200.224 is directly connected, FastEthernet1/0
     10.0.0.0/30 is subnetted, 2 subnets
C       10.10.13.0 is directly connected, FastEthernet0/0
C       10.10.23.0 is directly connected, FastEthernet0/1
S*   0.0.0.0/0 is directly connected, FastEthernet1/0
POD-X-R1#
```

sh ip route | begin Gateway sur R1 :

```
R1#sh ip route | begin Gateway
Gateway of last resort is 10.10.13.2 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.10.13.2, 01:05:55, GigabitEthernet0/0/1
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.10.13.0/30 is directly connected, GigabitEthernet0/0/1
L        10.10.13.1/32 is directly connected, GigabitEthernet0/0/1
O        10.10.23.0/30
           [110/2] via 172.16.30.3, 00:51:32, GigabitEthernet0/0/0.30
           [110/2] via 172.16.20.3, 00:51:32, GigabitEthernet0/0/0.20
           [110/2] via 172.16.10.3, 00:51:42, GigabitEthernet0/0/0.10
      172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C        172.16.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L        172.16.10.2/32 is directly connected, GigabitEthernet0/0/0.10
C        172.16.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L        172.16.20.2/32 is directly connected, GigabitEthernet0/0/0.20
C        172.16.30.0/24 is directly connected, GigabitEthernet0/0/0.30
L        172.16.30.2/32 is directly connected, GigabitEthernet0/0/0.30
R1#
```

sh ip route | begin Gateway sur R2 :

```
R2#sh ip route | begin Gateway
Gateway of last resort is 10.10.23.2 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.10.23.2, 01:05:24, GigabitEthernet0/0/1
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.10.13.0/30
           [110/2] via 172.16.30.2, 00:52:44, GigabitEthernet0/0/0.30
           [110/2] via 172.16.20.2, 00:52:44, GigabitEthernet0/0/0.20
           [110/2] via 172.16.10.2, 00:52:49, GigabitEthernet0/0/0.10
C        10.10.23.0/30 is directly connected, GigabitEthernet0/0/1
L        10.10.23.1/32 is directly connected, GigabitEthernet0/0/1
      172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C        172.16.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L        172.16.10.3/32 is directly connected, GigabitEthernet0/0/0.10
C        172.16.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L        172.16.20.3/32 is directly connected, GigabitEthernet0/0/0.20
C        172.16.30.0/24 is directly connected, GigabitEthernet0/0/0.30
L        172.16.30.3/32 is directly connected, GigabitEthernet0/0/0.30
R2#
```

sh ip route | begin Gateway sur POD-X-R1 :

```
POD-X-R1#sh ip route | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     172.16.0.0/24 is subnetted, 3 subnets
O       172.16.30.0 [110/11] via 10.10.23.1, 00:49:40, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:49:40, FastEthernet0/0
O       172.16.20.0 [110/11] via 10.10.23.1, 00:49:40, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:49:40, FastEthernet0/0
O       172.16.10.0 [110/11] via 10.10.23.1, 00:49:40, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:49:40, FastEthernet0/0
     209.165.200.0/27 is subnetted, 1 subnets
C       209.165.200.224 is directly connected, FastEthernet1/0
     10.0.0.0/30 is subnetted, 2 subnets
C       10.10.13.0 is directly connected, FastEthernet0/0
C       10.10.23.0 is directly connected, FastEthernet0/1
S*   0.0.0.0/0 is directly connected, FastEthernet1/0
POD-X-R1#
```

sh ipv6 route | begin C sur R1 :

```
R1#sh ipv6 route | begin C
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       a - Application
L   FF00::/8 [0/0]
     via Null0, receive
R1#
```

sh ip route static sur R1 :

```
R1#sh ip route static
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.10.13.2 to network 0.0.0.0

R1#
```

sh ip route static sur R2 :

```
R2#sh ip route static
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.10.23.2 to network 0.0.0.0

R2#
```

show ip protocols sur R1

```
R1#sh ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.10.13.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.13.0 0.0.0.3 area 0
    172.16.10.0 0.0.0.255 area 0
    172.16.20.0 0.0.0.255 area 0
    172.16.30.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.23.2           110      00:54:46
    172.16.10.3          110      01:07:21
  Distance: (default is 110)

R1#
```

show ip ospf neighbor sur R1

```
R1#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.10.3       1   FULL/DR         00:00:31    172.16.30.3     GigabitEthernet0/0/0.30
172.16.10.3       1   FULL/DR         00:00:36    172.16.20.3     GigabitEthernet0/0/0.20
172.16.10.3       1   FULL/DR         00:00:30    172.16.10.3     GigabitEthernet0/0/0.10
10.10.23.2        1   FULL/DR         00:00:36    10.10.13.2      GigabitEthernet0/0/1
R1#
```

show ip route ospf sur R1

```
R1#sh ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.10.13.2 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.10.13.2, 01:10:21, GigabitEthernet0/0/1
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.10.23.0/30
           [110/2] via 172.16.30.3, 00:55:58, GigabitEthernet0/0/0.30
           [110/2] via 172.16.20.3, 00:55:58, GigabitEthernet0/0/0.20
           [110/2] via 172.16.10.3, 00:56:08, GigabitEthernet0/0/0.10
R1#
```

show ip ospf interface sur R1

```
R1#sh ip ospf interface
GigabitEthernet0/0/0.30 is up, line protocol is up
  Internet Address 172.16.30.2/24, Area 0, Attached via Network Statement
  Process ID 1, Router ID 10.10.13.1, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 172.16.10.3, Interface address 172.16.30.3
  Backup Designated router (ID) 10.10.13.1, Interface address 172.16.30.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:03
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/4/4, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 2, maximum is 2
  Last flood scan time is 0 msec, maximum is 1 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 172.16.10.3  (Designated Router)
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0/0.20 is up, line protocol is up
  Internet Address 172.16.20.2/24, Area 0, Attached via Network Statement
  Process ID 1, Router ID 10.10.13.1, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 172.16.10.3, Interface address 172.16.20.3
  Backup Designated router (ID) 10.10.13.1, Interface address 172.16.20.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:05
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/3/3, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 2, maximum is 2
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 172.16.10.3  (Designated Router)
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0/0.10 is up, line protocol is up
  Internet Address 172.16.10.2/24, Area 0, Attached via Network Statement
  Process ID 1, Router ID 10.10.13.1, Network Type BROADCAST, Cost: 1
 --More--
```

```
  Last flood scan length is 2, maximum is 2
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 172.16.10.3  (Designated Router)
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0/0.10 is up, line protocol is up
  Internet Address 172.16.10.2/24, Area 0, Attached via Network Statement
  Process ID 1, Router ID 10.10.13.1, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 172.16.10.3, Interface address 172.16.10.3
  Backup Designated router (ID) 10.10.13.1, Interface address 172.16.10.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:04
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/2/2, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 2, maximum is 2
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 172.16.10.3  (Designated Router)
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0/1 is up, line protocol is up
  Internet Address 10.10.13.1/30, Area 0, Attached via Network Statement
  Process ID 1, Router ID 10.10.13.1, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 10.10.23.2, Interface address 10.10.13.2
  Backup Designated router (ID) 10.10.13.1, Interface address 10.10.13.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:02
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1/1, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 4, maximum is 4
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 10.10.23.2  (Designated Router)
  Suppress hello for 0 neighbor(s)
R1#
```

show ip ospf interface brief sur R1

```
R1#sh ip ospf interface br
Interface     PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Gi0/0/0.30    1     0               172.16.30.2/24     1     BDR   1/1
Gi0/0/0.20    1     0               172.16.20.2/24     1     BDR   1/1
Gi0/0/0.10    1     0               172.16.10.2/24     1     BDR   1/1
Gi0/0/1       1     0               10.10.13.1/30      1     BDR   1/1
R1#
```

show ip route ospf | begin 10 sur R1

```
R1#sh ip route ospf | begin 10
Gateway of last resort is 10.10.13.2 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.10.13.2, 01:12:40, GigabitEthernet0/0/1
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.10.23.0/30
           [110/2] via 172.16.30.3, 00:58:17, GigabitEthernet0/0/0.30
           [110/2] via 172.16.20.3, 00:58:17, GigabitEthernet0/0/0.20
           [110/2] via 172.16.10.3, 00:58:27, GigabitEthernet0/0/0.10
R1#
```

show ip route ospf sur POD-X-R1

```
POD-X-R1#sh ip route ospf
     172.16.0.0/24 is subnetted, 3 subnets
O       172.16.30.0 [110/11] via 10.10.23.1, 00:58:48, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:58:48, FastEthernet0/0
O       172.16.20.0 [110/11] via 10.10.23.1, 00:58:48, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:58:48, FastEthernet0/0
O       172.16.10.0 [110/11] via 10.10.23.1, 00:58:48, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:58:48, FastEthernet0/0
POD-X-R1#
```

traceroute 172.16.10.1 depuis pc1 :

```
root@poda-1:~# traceroute 172.16.10.1
traceroute to 172.16.10.1 (172.16.10.1), 30 hops max, 60 byte packets
 1  172.16.10.2 (172.16.10.2)  0.627 ms * *
root@poda-1:~#
```

traceroute 209.165.200.225 depuis pc1 :

```
root@poda-1:~# traceroute 209.165.200.225
traceroute to 209.165.200.225 (209.165.200.225), 30 hops max, 60 byte packets
 1  172.16.10.2 (172.16.10.2)  0.254 ms  0.432 ms  0.403 ms
 2  10.10.13.2 (10.10.13.2)  16.007 ms  26.141 ms  36.282 ms
root@poda-1:~#
```

traceroute 172.16.20.3 depuis pc1 :

```
root@poda-1:~# traceroute 172.16.20.3
traceroute to 172.16.20.3 (172.16.20.3), 30 hops max, 60 byte packets
 1  172.16.10.2 (172.16.10.2)  0.264 ms  0.452 ms  0.441 ms
 2  172.16.20.3 (172.16.20.3)  1.223 ms * *
root@poda-1:~#
```

ping depuis PC1 vers la passerelle virtuelle avant enlever le câble entre R1 et S1

```
root@poda-1:~# ping 172.16.10.1
PING 172.16.10.1 (172.16.10.1) 56(84) bytes of data.
64 bytes from 172.16.10.1: icmp_seq=1 ttl=255 time=0.645 ms
64 bytes from 172.16.10.1: icmp_seq=2 ttl=255 time=0.572 ms
64 bytes from 172.16.10.1: icmp_seq=3 ttl=255 time=0.670 ms
64 bytes from 172.16.10.1: icmp_seq=4 ttl=255 time=0.669 ms
64 bytes from 172.16.10.1: icmp_seq=5 ttl=255 time=0.639 ms
64 bytes from 172.16.10.1: icmp_seq=6 ttl=255 time=0.731 ms
64 bytes from 172.16.10.1: icmp_seq=7 ttl=255 time=0.725 ms
64 bytes from 172.16.10.1: icmp_seq=8 ttl=255 time=0.761 ms
64 bytes from 172.16.10.1: icmp_seq=9 ttl=255 time=0.711 ms
```

ping depuis PC1 vers la passerelle virtuelle avant et après enlever le câble entre R1 et S1

```
root@poda-1:~# ping 172.16.10.1
PING 172.16.10.1 (172.16.10.1) 56(84) bytes of data.
64 bytes from 172.16.10.1: icmp_seq=1 ttl=255 time=0.621 ms
64 bytes from 172.16.10.1: icmp_seq=2 ttl=255 time=0.618 ms
64 bytes from 172.16.10.1: icmp_seq=3 ttl=255 time=0.623 ms
64 bytes from 172.16.10.1: icmp_seq=4 ttl=255 time=0.799 ms
64 bytes from 172.16.10.1: icmp_seq=5 ttl=255 time=0.664 ms
64 bytes from 172.16.10.1: icmp_seq=6 ttl=255 time=0.623 ms
64 bytes from 172.16.10.1: icmp_seq=7 ttl=255 time=0.658 ms
64 bytes from 172.16.10.1: icmp_seq=8 ttl=255 time=0.579 ms
64 bytes from 172.16.10.1: icmp_seq=9 ttl=255 time=0.622 ms
64 bytes from 172.16.10.1: icmp_seq=10 ttl=255 time=0.783 ms
64 bytes from 172.16.10.1: icmp_seq=11 ttl=255 time=0.627 ms
64 bytes from 172.16.10.1: icmp_seq=12 ttl=255 time=0.687 ms
64 bytes from 172.16.10.1: icmp_seq=22 ttl=255 time=0.861 ms
64 bytes from 172.16.10.1: icmp_seq=23 ttl=255 time=0.874 ms
64 bytes from 172.16.10.1: icmp_seq=24 ttl=255 time=1.09 ms
64 bytes from 172.16.10.1: icmp_seq=25 ttl=255 time=1.12 ms
64 bytes from 172.16.10.1: icmp_seq=26 ttl=255 time=1.00 ms
64 bytes from 172.16.10.1: icmp_seq=27 ttl=255 time=1.16 ms
64 bytes from 172.16.10.1: icmp_seq=28 ttl=255 time=1.03 ms
64 bytes from 172.16.10.1: icmp_seq=29 ttl=255 time=1.09 ms
64 bytes from 172.16.10.1: icmp_seq=30 ttl=255 time=1.06 ms
64 bytes from 172.16.10.1: icmp_seq=31 ttl=255 time=1.09 ms
64 bytes from 172.16.10.1: icmp_seq=32 ttl=255 time=0.968 ms
64 bytes from 172.16.10.1: icmp_seq=33 ttl=255 time=1.05 ms
64 bytes from 172.16.10.1: icmp_seq=34 ttl=255 time=1.01 ms
64 bytes from 172.16.10.1: icmp_seq=35 ttl=255 time=1.05 ms
64 bytes from 172.16.10.1: icmp_seq=36 ttl=255 time=1.14 ms
64 bytes from 172.16.10.1: icmp_seq=37 ttl=255 time=1.11 ms
64 bytes from 172.16.10.1: icmp_seq=38 ttl=255 time=0.984 ms
64 bytes from 172.16.10.1: icmp_seq=39 ttl=255 time=1.10 ms
64 bytes from 172.16.10.1: icmp_seq=40 ttl=255 time=1.04 ms
64 bytes from 172.16.10.1: icmp_seq=41 ttl=255 time=5.38 ms
64 bytes from 172.16.10.1: icmp_seq=42 ttl=255 time=1.10 ms
64 bytes from 172.16.10.1: icmp_seq=43 ttl=255 time=1.04 ms
64 bytes from 172.16.10.1: icmp_seq=44 ttl=255 time=1.05 ms
^C
--- 172.16.10.1 ping statistics ---
44 packets transmitted, 35 received, 20.4545% packet loss, time 43568ms
rtt min/avg/max/mdev = 0.579/1.037/5.376/0.769 ms
root@poda-1:~#
```

traceroute 172.16.10.1 sur pc1 après reconnecter le câble

le routeur **R1** (`172.16.10.2`) a automatiquement repris son rôle de passerelle active. Ce résultat démontre que l'architecture est non seulement résiliente face aux pannes, mais qu'elle est aussi capable de **revenir d'elle-même à son état nominal** sans intervention humaine, garantissant ainsi une gestion optimale des flux de données.

```
root@poda-1:~# traceroute 172.16.10.1
traceroute to 172.16.10.1 (172.16.10.1), 30 hops max, 60 byte packets
 1  172.16.10.2 (172.16.10.2)  0.634 ms * *
root@poda-1:~#
```

show ip nat translations sur POD-X-R1

```
POD-X-R1#sh ip nat translations
Pro Inside global      Inside local       Outside local      Outside global
udp 209.165.200.225:43453 172.16.10.22:43453 8.8.8.8:53      8.8.8.8:53
udp 209.165.200.225:56011 172.16.10.22:56011 8.8.8.8:53      8.8.8.8:53
POD-X-R1#
```

show ip route sur POD-X-R1

```
POD-X-R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     172.16.0.0/24 is subnetted, 3 subnets
O       172.16.30.0 [110/11] via 10.10.23.1, 00:01:27, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:01:27, FastEthernet0/0
O       172.16.20.0 [110/11] via 10.10.23.1, 00:01:27, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:01:27, FastEthernet0/0
O       172.16.10.0 [110/11] via 10.10.23.1, 00:01:27, FastEthernet0/1
                    [110/11] via 10.10.13.1, 00:01:27, FastEthernet0/0
     209.165.200.0/27 is subnetted, 1 subnets
C       209.165.200.224 is directly connected, FastEthernet1/0
     10.0.0.0/30 is subnetted, 2 subnets
C       10.10.13.0 is directly connected, FastEthernet0/0
C       10.10.23.0 is directly connected, FastEthernet0/1
S*   0.0.0.0/0 is directly connected, FastEthernet1/0
POD-X-R1#
```

ping depuis POD-X-R2 vers l'interface Bord 209.165.200.225 succes

Valide la connectivité physique et IP du lien Série ou Ethernet entre l'ISP et l'entreprise.

```
POD-X-R2#ping 209.165.200.225

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.225, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/20/28 ms
POD-X-R2#
```

ping depuis POD-X-R2 vers VLAN 10 172.16.10.10 echec

Les adresses privées ne sont pas routables sur Internet. Seule l'IP NATée est joignable.

```
POD-X-R2#ping 172.16.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
POD-X-R2#
```

show ip inter br sur POD-X-R2

```
POD-X-R2#sh ip int br
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            209.165.200.226 YES manual up                    up
Serial0/0                  unassigned      YES unset  administratively down down
FastEthernet0/1            unassigned      YES unset  administratively down down
Serial0/1                  unassigned      YES unset  administratively down down
FastEthernet1/0            unassigned      YES unset  administratively down down
FastEthernet2/0            unassigned      YES unset  administratively down down
FastEthernet3/0            unassigned      YES unset  administratively down down
FastEthernet4/0            unassigned      YES unset  administratively down down
Loopback0                  209.165.100.1   YES manual up                    up
POD-X-R2#
```

show ip route sur POD-X-R2

```
POD-X-R2#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     209.165.200.0/27 is subnetted, 1 subnets
C       209.165.200.224 is directly connected, FastEthernet0/0
     209.165.100.0/32 is subnetted, 1 subnets
C       209.165.100.1 is directly connected, Loopback0
POD-X-R2#
```

Ping depuis pc1 vers 209.165.100.1

```
root@poda-1:~# ping 209.165.100.1
PING 209.165.100.1 (209.165.100.1) 56(84) bytes of data.
64 bytes from 209.165.100.1: icmp_seq=1 ttl=253 time=30.1 ms
64 bytes from 209.165.100.1: icmp_seq=2 ttl=253 time=29.6 ms
64 bytes from 209.165.100.1: icmp_seq=3 ttl=253 time=30.9 ms
64 bytes from 209.165.100.1: icmp_seq=4 ttl=253 time=31.8 ms
64 bytes from 209.165.100.1: icmp_seq=5 ttl=253 time=21.7 ms
64 bytes from 209.165.100.1: icmp_seq=6 ttl=253 time=21.6 ms
64 bytes from 209.165.100.1: icmp_seq=7 ttl=253 time=21.7 ms
64 bytes from 209.165.100.1: icmp_seq=8 ttl=253 time=21.7 ms
64 bytes from 209.165.100.1: icmp_seq=9 ttl=253 time=32.4 ms
64 bytes from 209.165.100.1: icmp_seq=10 ttl=253 time=12.8 ms
64 bytes from 209.165.100.1: icmp_seq=11 ttl=253 time=22.5 ms
64 bytes from 209.165.100.1: icmp_seq=12 ttl=253 time=22.4 ms
64 bytes from 209.165.100.1: icmp_seq=13 ttl=253 time=23.0 ms
64 bytes from 209.165.100.1: icmp_seq=14 ttl=253 time=23.5 ms
```

sh ip nat translations sur POD-X-R1

```
POD-X-R1#sh ip nat translations
Pro Inside global      Inside local       Outside local      Outside global
icmp 209.165.200.225:11392 172.16.10.22:11392 209.165.100.1:11392 209.165.100.1:11392
--- 209.165.200.227    172.16.20.10       ---                ---
POD-X-R1#
```

Ping depuis POD-X-R2 vers 209.165.200.227

```
POD-X-R2#ping 209.165.200.227

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.227, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 20/25/32 ms
```

Ping depuis POD-X-R2 vers 209.165.200.225

```
POD-X-R2#ping 209.165.200.225

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.225, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/11/24 ms
```